

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1279

Cold Spring Rd 138 kV
December 31, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1279. The Interconnection Customer proposes a 362.3 MW solar photovoltaic facility interconnecting at the Cold Spring Rd 138 kV substation in the Virginia area. The request was received on September 01, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 437 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$63,288,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Anna - Cold Spring Rd 138.0 kV	Fluvanna P.S - Louisa Generation Facility 345.0 kV	239.8	211	84.6
Apple Data Ctr Pv - Cold Spring Rd 138.0 kV	Cold Spring Rd - N Anna 138.0 kV line	172.7	210	100.0
Fern Vly Substation - Apple Data Center Pv 138.0 kV	Cold Spring Rd - N Anna 138.0 kV	154.0	223	100.0
Cabrillo - Fern Valley 138.0 kV line	Cold Spring Rd - N Anna 138.0 kV	149.1	227	100.0
Abbeville - Cabrillo 138.0 kV Line	Cold Spring Rd - N Anna 138.0 kV Line	140.4	235	100.0
Instituto Tecnologico De Ponce - Abbeville 138.0 kV Line	N Anna - Cold Spring d 138.0 kV Line	137.1	236	100.0
Golden Hills - Instituto Tecnologico De Ponce 138.0 kV	Cold Spring Rd - N Anna 138.0 kV	133.8	238	100.0
Golden Hills - Willow Crk 138.0 kV line	Cold Spring Rd - North Anna 138.0 kV	129.6	240	100.0
Willow Crk - Breemo Bluff 138.0 kV	Cold Spring Rd - N Ana 138.0 kV	123.7	245	100.0
Chancellor - Ni River Dp 138.0 kV Line	N Anna 345/138 kV Transformer Bank	117.4	224	64.7
Story City - Chancellor 138.0 kV L	North Anna 345/138 kV Autotransformer	109.8	209	64.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Anna - Cold Spring Rd 138.0 kV	Rebuild line with higher-capacity conductor	\$3,188,000	98% (\$3,127,000)
Apple Data Ctr Pv - Cold Spring Rd 138.0 kV	Rebuild line with higher-capacity conductor	\$1,968,000	100% (\$1,968,000)
Fern Vly Substation - Apple Data Center Pv 138.0 kV	Rebuild line with higher-capacity conductor	\$8,580,000	100% (\$8,580,000)
Cabrillo - Fern Valley 138.0 kV line	Rebuild line with higher-capacity conductor	\$13,089,000	100% (\$13,089,000)
Abbeville - Cabrillo 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,014,000	100% (\$6,014,000)
Instituto Tecnologico De Ponce - Abbeville 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,916,000	100% (\$5,916,000)
Golden Hills - Instituto Tecnologico De Ponce 138.0 kV	Rebuild line with higher-capacity conductor	\$8,963,000	100% (\$8,963,000)
Golden Hills - Willow Crk 138.0 kV line	Rebuild line with higher-capacity conductor	\$7,730,000	100% (\$7,730,000)
Willow Creek - Brmo Bluff 138.0 kV	Rebuild line with higher-capacity conductor	\$4,607,000	100% (\$4,607,000)
Chancellor - Ni River Dp 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,924,000	84% (\$2,464,000)
Chancellor - Story City 138.0 kV	Reconductor line	\$985,000	84% (\$830,000)
	Total allocated to project		\$63,288,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.